

input
features

output
target

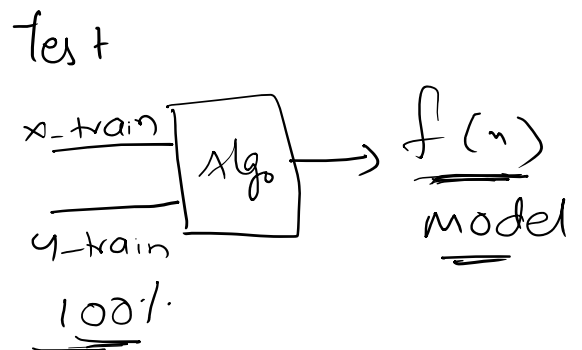
x_1	x_2	x_3	...		y
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$$y = \text{max}(0)$$

$$y = \text{max}(w_0 + w_1x_1 + w_2x_2 + \dots + w_nx_n)$$

100 samples						X	Y
	x_1	x_2	x_3	...	x_n		y
80%	x_{train}						y_{train}
20%	<u>x_{test}</u>						<u>y_{test}</u>

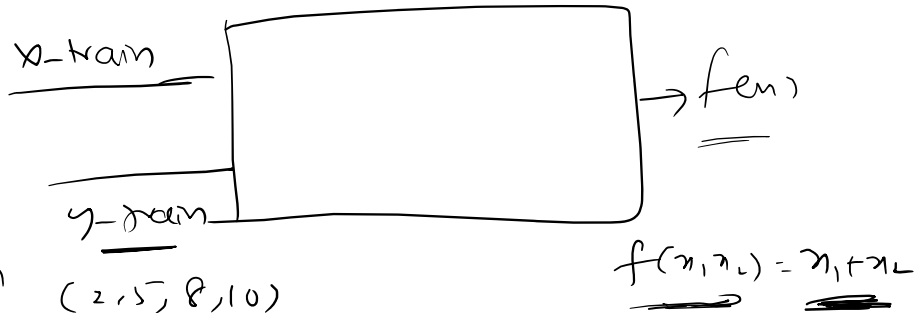
Split $\rightarrow$ 80% 20%
70% 30%
60% 40%
50% 50%



$f(x_{\text{test}})$ = Answers (prediction)

	x_1	x_2	y
80% x_{train}	1	1	2
	3	2	5
	6	2	8
	9	1	10
20% x_{test}	10	1	11
	1	5	6

(1,1) (3,2) (6,2) (9,1)

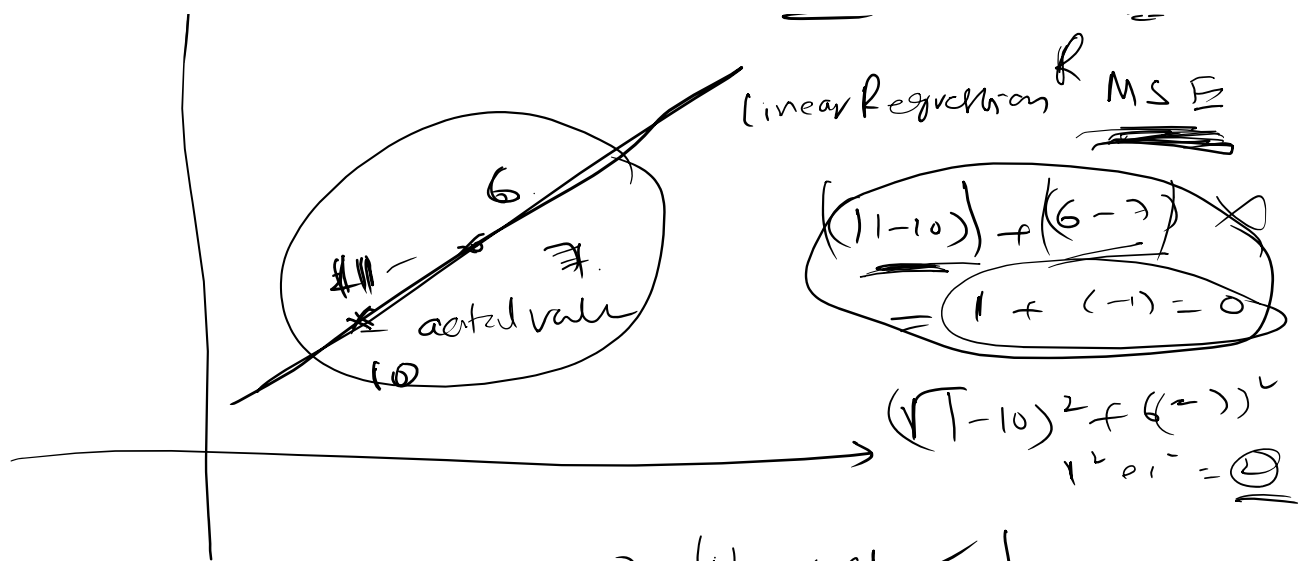


$f(10, 1) = 11$
 $f(1, 5) = 6$ we get here model

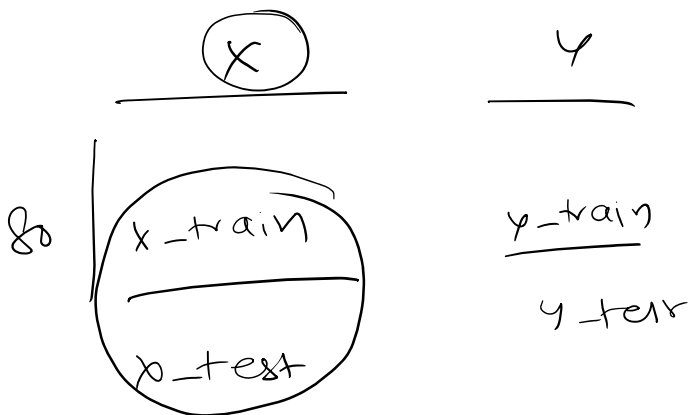
100%

Accuracy

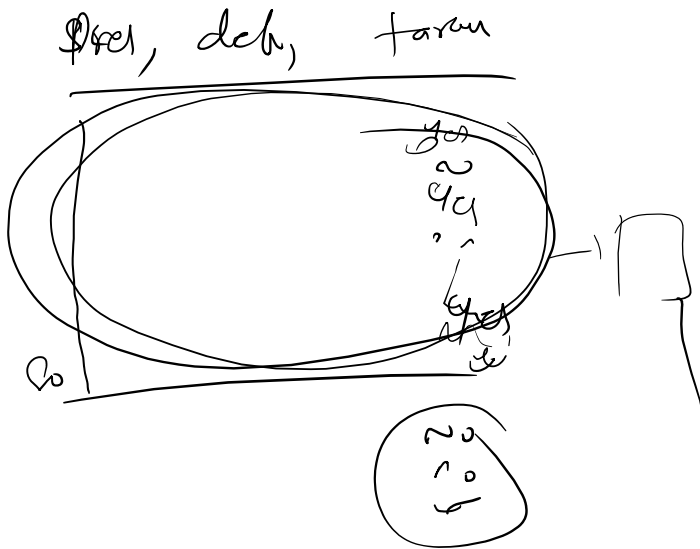
Linear Regression MSE



- ① import all the required libraries ✓
- ② read the data into dataframe ✓
- ③ preprocess the data ✓
- ④ define $x = ?$ $y = ?$ ✓
- ⑤ split x and y into train and test
- ⑥ create an object of algorithm
- ⑦ fit the data to the object
- ⑧ find the accuracy



x-train, x-test, y-train, y-test = (x, y, test = 0.2)



① pattern



② multiple

③ data/synthesis

Supervised Learning

→ Regression ✓
→ classification

classification

is data

SL SW PL PW (specific)

target	train
20	yes
21	
22	
23	



SL SW PL PW

KNW

class like

CPeKID
fctos
ver:com
vmin

1

class like